
PHT 310 / PSY 111: Introduction to Psychology

Units: 2 | Lecture Hours: 30

LECTURE 1: MEANING, NATURE, AND AIMS OF PSYCHOLOGY

Objectives:

- Define psychology and its scope.
- Differentiate behaviour from mental processes.
- Explain the aims of psychology.
- Discuss relevance to physiotherapy and rehabilitation programs.

Content:

- **Definition:** Psychology = Psyche (mind) + Logos (study) → scientific study of behaviour and mental processes.
- **Behaviour:** Observable actions (walking, speaking, crying, performing exercises).
- **Mental Processes:** Internal experiences (thinking, memory, emotion, perception, motivation, intelligence).

Nature of Psychology:

- **Scientific:** Relies on observation, experimentation, evidence.
- **Biological & Social:** Interplay between brain-body systems and culture/environment.
- **Applied & Theoretical:** Generates knowledge and practical solutions.

Aims:

1. **Description:** Observe behaviour (e.g., patient grimacing during pain).

2. **Explanation:** Understand causes (fear of movement post-injury).
3. **Prediction:** Anticipate behaviour (exercise adherence).
4. **Control/Modification:** Encourage positive behaviour (rewards for exercise).
5. **Application:** Improve health, rehabilitation, and functional outcomes.

Physiotherapy Relevance:

- Understanding pain perception differences.
- Emotional responses affecting motivation.
- Tailoring rehabilitation strategies to cognitive and emotional needs.

Criminology Relevance:

- Understanding behavioural patterns in offenders.
- Assessing risk and designing rehabilitation programs for juvenile offenders.

Example: Adjusting exercise intensity for a patient anxious about re-injury; implementing structured behaviour reinforcement for at-risk youth.

References: Myers (2023); Ciccarelli & White (2017).

LECTURE 2: HISTORICAL AND PHILOSOPHICAL ANTECEDENTS

Objectives:

- Identify key philosophical contributors.
- Explain evolution from philosophy to modern psychology.
- Apply historical insights to physiotherapy and rehabilitation.

Content:

- **Philosophical Foundations:**
 - Socrates & Plato – mind-body dualism, innate knowledge.
 - Aristotle – empirical observation, learning from experience.
 - Descartes – reflex actions, dualism.
 - Locke – Tabula Rasa (mind as blank slate).
- **Scientific Contributions:**
 - Weber & Fechner – psychophysics.
 - Helmholtz – nerve conduction, reaction time studies.
 - Darwin – evolutionary adaptation of behaviour.
- **Modern Psychology:** Wundt (1879) – experimental lab; studied consciousness via introspection.

Physiotherapy Relevance:

- Sensory perception changes after injury.
- Adaptive learning principles for motor skills rehabilitation.

Criminology Relevance:

- Evolutionary psychology explains aggression, antisocial behaviour.
- Empirical methods guide evidence-based rehabilitation programs.

Example: Customizing exercises for patients with neuropathy; structured intervention for adolescents with aggressive tendencies.

References: Schultz & Schultz (2016); Hergenhahn (2009).

LECTURE 3: EARLY SCHOOLS OF PSYCHOLOGY

Objectives:

- Explain early schools and contributions.
- Discuss limitations and clinical relevance.

Content:

- **Structuralism (Wundt, Titchener):** Introspection to study consciousness.
- **Functionalism (James):** Behaviour as adaptation.
- **Behaviourism (Watson, Skinner):** Observable behaviour; classical & operant conditioning.
- **Gestalt Psychology (Wertheimer, Köhler, Koffka):** Whole > sum of parts; perception patterns.
- **Psychoanalysis (Freud):** Id, Ego, Superego; unconscious processes.

Physiotherapy Relevance:

- Behavioural reinforcement improves exercise compliance.
- Gestalt principles enhance motor learning.
- Understanding emotional trauma post-injury.

Criminology Relevance:

- Behaviourist techniques in offender rehabilitation (reward/punishment).
- Psychoanalytic principles to understand antisocial behaviour and treatment approaches.

Example: Rewarding patients for completing exercises; cognitive-behavioural programs for juvenile offenders.

References: Freud (1923); James (1890).

LECTURE 4: CONTEMPORARY SCHOOLS OF PSYCHOLOGY

Objectives:

- Describe contemporary schools.
- Apply concepts to patient-centred care and offender rehabilitation.

Content:

- **Humanistic (Maslow, Rogers):** Self-actualization; patient-centred approach.
- **Cognitive Psychology:** Problem-solving, memory strategies.
- **Biological Psychology:** Neurotransmitters, brain function, genetics.
- **Evolutionary Psychology:** Behavioural adaptations.
- **Cultural & Positive Psychology:** Cultural influence, resilience, well-being.

Physiotherapy Relevance:

- Designing patient-specific rehabilitation programs.
- Encouraging self-efficacy and motivation.

Criminology Relevance:

- Humanistic approaches in offender rehabilitation.
- Cognitive strategies for behaviour modification and reintegration.

Example: Stepwise goal-setting for stroke rehabilitation; CBT-based rehabilitation for youth offenders.

References: Maslow (1943); Rogers (1951).

LECTURE 5: FIELDS OF PSYCHOLOGY

Objectives:

- Identify major fields.
- Explain relevance to healthcare, physiotherapy, and criminology.

Content:

- Fields: Clinical, counselling, developmental, educational, social, industrial, neuropsychology, health, forensic psychology.

Physiotherapy Relevance:

- Health psychology for chronic illness adjustment.
- Neuropsychology aids neurological recovery.

Criminology Relevance:

- Forensic psychology supports assessment, risk management, and rehabilitation of offenders.

Example: Cognitive rehabilitation for stroke recovery; structured intervention for juvenile offenders.

References: Sarafino (2012).

LECTURE 6: PSYCHOLOGY AS A SCIENTIFIC DISCIPLINE

Objectives:

- Explain principles of scientific psychology.
- Apply evidence-based practice in physiotherapy and rehabilitation programs.

Content:

- Principles: Empiricism, objectivity, determinism, skepticism, testability.
- Evidence-based psychology improves clinical decision-making.

Physiotherapy Example: Validated scales to monitor pain, functional progress.

Criminology Example: Structured assessments for offender rehabilitation planning.

References: Stanovich (2013).

LECTURE 7: METHODS OF PSYCHOLOGICAL INVESTIGATION

Objectives:

- Identify research methods.
- Apply methods to physiotherapy and criminology.

Content:

- Experiment, observation, case study, survey, correlational research.

Physiotherapy Application: Studying motor learning, adherence, rehabilitation outcomes.

Criminology Application: Studying offender behaviour, interventions, risk factors.

Example: Controlled trial on physiotherapy interventions; evaluation of anger management program for juvenile offenders.

References: Creswell (2018).

LECTURE 8: PSYCHOLOGY AND OTHER DISCIPLINES

Objectives:

- Explain interdisciplinary connections.
- Apply interdisciplinary knowledge in healthcare and rehabilitation.

Content:

- Links: biology, sociology, anthropology, medicine, education, criminology.

Physiotherapy Relevance: Holistic care; patient background, social support, adherence.

Criminology Relevance: Social context informs offender assessment and rehabilitation.

Example: Family support in stroke recovery; community-based rehabilitation for ex-offenders.

LECTURE 9: ISSUES AND CONTROVERSIES

Objectives:

1. Discuss key psychological debates:
 - **Nature vs Nurture**
 - **Free Will vs Determinism**
2. Apply ethical principles in professional practice.

1. Psychological Debates

Nature vs Nurture:

- **Nature:** Behavior is determined by genetics, biology.
- **Nurture:** Behavior is shaped by environment, upbringing, experiences.
- **Implication:** Understanding origin of behavior helps in designing interventions.

Free Will vs Determinism:

- **Free Will:** Individuals can make independent choices.
- **Determinism:** Behavior is influenced or controlled by external/internal factors (e.g., biology, environment).
- **Implication:** Guides ethical responsibility in therapy or rehabilitation programs.

2. Ethical Principles in Practice

- **Informed Consent:** Ensure participants understand procedures, risks, and benefits.
- **Autonomy:** Respect patient or client decisions regarding their care.
- **Cultural Sensitivity:** Adapt approaches to align with cultural beliefs and values.

- **Beneficence & Non-Maleficence:** Promote good outcomes while avoiding harm.
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3. Relevance in Different Fields

Physiotherapy:

- Ensuring patients voluntarily participate in treatment programs.
- Designing therapy plans respecting individual autonomy and cultural norms.
- Example: Voluntary participation in experimental therapy.

Criminology:

- Designing ethical rehabilitation programs for offenders.
- Balancing punishment with supportive interventions to reduce recidivism.
- Example: Structured voluntary programs for juvenile rehabilitation.

4. Key Takeaways:

- Ethical practice bridges psychological theory and professional application.
 - Awareness of debates like **nature vs nurture** and **free will vs determinism** informs treatment and rehabilitation strategies.
 - Voluntary participation and respect for autonomy are central across disciplines.
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LECTURE 10: INTELLIGENCE AND MEASUREMENT

Objectives

- Explain major intelligence theories
- Describe IQ measurement and applications

1. Intelligence Theories

- **Spearman – g Factor:** Single general intelligence influencing all tasks
- **Thurstone – Primary Abilities:** Several independent abilities (verbal, numerical, spatial, etc.)
- **Gardner – Multiple Intelligences:** Diverse intelligences:
 - Linguistic, Logical-Mathematical, Musical
 - Bodily-Kinesthetic, Interpersonal, Intrapersonal, Naturalistic

2. IQ Measurement

- Measures cognitive functioning: reasoning, problem-solving, memory
- Helps identify **strengths & weaknesses**

3. Applications

Physiotherapy:

- Tailor rehab programs to cognitive abilities
- Adjust instructions for learning/comprehension challenges

Criminology:

- Assess cognitive strengths and limitations of offenders
- Plan **effective, individualized rehabilitation programs**

4. Key Takeaways

- Intelligence is multi-dimensional; no one theory explains everything
- IQ assessment informs practical interventions
- Ethical use: **support and guide**, don't limit or label individuals

LECTURE 11: LEARNING AND DEVELOPMENT

Objectives

- Explain major **learning theories**
- Describe **stages of human development**
- Apply concepts in physiotherapy and criminology

1. Learning Theories

- **Classical Conditioning (Pavlov):** Learning through association (e.g., pairing stimuli to elicit a response)
- **Operant Conditioning (Skinner):** Learning through consequences (reinforcement & punishment)
- **Observational Learning (Bandura):** Learning by watching others' behavior and outcomes

2. Stages of Development

- **Physical Development:** Growth of body and motor skills
- **Cognitive Development:** Thinking, problem-solving, reasoning
- **Emotional Development:** Understanding and regulating emotions
- **Social Development:** Interaction, communication, relationships

3. Applications

Physiotherapy:

- Motor skill acquisition through repeated practice (classical & operant conditioning)
- Enhancing exercise adherence using reinforcement strategies
- Observational learning: demonstrating exercises for patients

Criminology:

- Behaviour modification programs using reinforcement and observation
- Developing pro-social behaviors in offenders
- Tailoring interventions according to cognitive & emotional development

4. Key Takeaways

- Learning theories explain **how behavior is acquired or changed**
- Understanding development stages helps **design age-appropriate interventions**
- Ethical application ensures **supportive, effective rehabilitation**

LECTURE 12: PERCEPTION AND PERSON PERCEPTION

Objectives

- Explain **perception processes**
- Describe **person perception**

1. Perception Processes

- **Sensation → Organization → Interpretation**
- **Gestalt Principles:**
 - **Proximity:** Objects close together perceived as a group
 - **Similarity:** Similar items grouped together
 - **Closure:** Incomplete shapes seen as complete
 - **Figure-Ground:** Distinguishing object from background

2. Applications

Physiotherapy:

- Improve **communication** with patients
- Optimize **environment setup** for therapy sessions
- Enhance **patient engagement** through visual and spatial cues

Criminology:

- Understand **offender perception and biases**
- Interpret **social cues and interactions** in assessment or rehabilitation

- Design interventions considering **how individuals perceive situations**

3. Key Takeaways

- Perception is an **active process** of organizing and interpreting sensory information
- Gestalt principles help **predict and influence perception**
- Understanding perception enhances **effective communication and behavior management**

LECTURE 13: IMPRESSION FORMATION AND ATTRIBUTION

Objectives

- Explain common **cognitive and social biases**:
 - **Halo Effect**
 - **Fundamental Attribution Error**
 - **Self-Serving Bias**

1. Common Biases

- **Halo Effect:** Judging a person based on one positive or negative trait
- **Fundamental Attribution Error:** Overemphasizing personal traits and underestimating situational factors in explaining others' behavior
- **Self-Serving Bias:** Attributing successes to oneself and failures to external factors

2. Applications

Physiotherapy:

- Improve **clinical judgment** by recognizing bias in patient assessment
- Enhance **patient rapport** and trust

- Reduce **misinterpretation of patient behavior or progress**

Criminology:

- Reduce **bias in offender evaluation**
- Ensure **fair and effective rehabilitation planning**
- Support evidence-based **decision-making**

3. Key Takeaways

- Biases influence **perception, judgment, and decision-making**
 - Awareness of biases improves **professional practice** in both physiotherapy and criminology
 - Ethical practice requires **recognizing and mitigating bias** to support better outcomes
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LECTURE 14: SOCIALIZATION AND JUVENILE DELINQUENCY

Objectives

- Explain **agents of socialization**
- Discuss **causes and prevention of delinquency**

1. Agents of Socialization

- **Family:** Shapes values, behavior, and norms
- **School:** Academic, social, and moral development
- **Media:** Influences attitudes and behavior
- **Religion:** Provides moral guidance and social norms

2. Causes of Delinquency

- Family instability or poor supervision
- Peer influence and group pressure
- Poverty and lack of resources
- Substance abuse (drugs, alcohol)
- School failure or disengagement

- Exposure to media violence

3. Applications

Physiotherapy:

- Encourage **social support networks** for patients
- Promote **structured physical activity** to reduce risk behaviors
- Exercise to enhance **discipline and self-regulation**

Criminology:

- Prevention strategies targeting **risk factors**
- Rehabilitation programs tailored to **social/environmental influences**
- Community-based programs to reduce delinquency

4. Key Takeaways

- Socialization strongly influences **behavior and moral development**
- Understanding causes of delinquency helps in **prevention and rehabilitation**
- Effective interventions address **family, school, peer, and community factors**

LECTURE 15: BEHAVIOUR MODIFICATION AND AGGRESSION

Objectives

- Explain **behaviour modification techniques**
- Describe **theories of aggression**

1. Behaviour Modification

- **Reinforcement:** Reward desired behavior
- **Shaping:** Gradually guide behavior toward target

- **Extinction:** Reduce undesired behavior by removing reinforcement
- **Modelling:** Learn by observing others

2. Theories of Aggression

- **Biological:** Genes, brain, hormones
- **Frustration-Aggression:** Aggression when goals are blocked
- **Social Learning:** Learned by observing and imitating others

3. Applications

Physiotherapy:

- Manage **patient frustration**
- Improve **compliance with exercises**
- Reduce **aggressive/disruptive responses**

Criminology:

- Anger management programs for offenders
- Social skills training to reduce recidivism
- Model **pro-social behaviors** in rehabilitation

4. Key Takeaways

- Behaviour modification **shapes positive behavior**
- Aggression arises from **biological, situational, and social factors**
- Ethical practice supports **safe management of aggression**

LECTURE 16: PSYCHOLOGY, CRIMINOLOGY, AND REHABILITATION

Objectives

- Integrate **psychology into offender and victim rehabilitation**

1. Criminology Overview

- Study of **crime, its causes, prevention, and rehabilitation**
- **Theories of crime:**
 - Psychoanalytic, Behavioural, Cognitive, Biological, Social/Environmental

2. Personality & Mental Health Factors

- Traits influencing offending: **impulsivity, aggression, antisocial personality, substance use**
- Juvenile delinquency: causes, prevention, rehabilitation

3. Victim Psychology

- Psychological impacts of crime: **PTSD, depression, physical disability**
- Rehabilitation must address **mental, emotional, and physical needs**

4. Rehabilitation & Behaviour Modification

- Techniques:
 - **CBT** – cognitive restructuring for maladaptive thoughts
 - **Anger management** – reducing aggression
 - **Social skills training** – improving interpersonal functioning
 - **Vocational training** – reintegration into society

Example:

- 19-year-old male with **spinal injury from cult violence** → combine **psychological support, social reintegration, and physiotherapy** for optimal recovery

5. Physiotherapy Relevance

- Injury rehabilitation, neurological recovery, functional restoration
- Enhances recovery outcomes by **integrating psychological strategies**

6. Key Takeaways

- Rehabilitation is **holistic**, addressing psychological, social, and physical factors
 - Behaviour modification and psychological interventions **support long-term recovery and reintegration**
 - Effective rehabilitation relies on **interdisciplinary collaboration**
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PTSD (Post-Traumatic Stress Disorder) – Summary

- **Definition:** Mental health condition after **trauma**; causes distressing memories, emotional disturbance, and functional impairment.
 - **Key Symptoms:**
 - **Intrusion:** Flashbacks, nightmares
 - **Avoidance:** Steering clear of reminders
 - **Negative Mood:** Guilt, depression, emotional numbness
 - **Arousal:** Irritability, hypervigilance, sleep issues
 - **Causes/Risk Factors:** Trauma exposure, genetics, prior mental health issues, lack of support
 - **Applications:**
 - **Physiotherapy:** Trauma-informed care; reduce stress during rehab
 - **Criminology:** Recognize PTSD in victims/offenders; integrate into rehabilitation
 - **Management:** CBT, EMDR, medication, social support
 - **Takeaway:** Early recognition and **holistic, interdisciplinary care** improve recovery outcomes
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Psychological Interventions: CBT & EMDR

1. Cognitive Behavioral Therapy (CBT)

- **Definition:** Structured, time-limited therapy focused on **changing maladaptive thoughts and behaviors**
- **How it Works:**
 - Identify **negative thought patterns**
 - Challenge and **restructure thoughts**
 - Encourage **positive coping strategies and behavioral change**
- **Applications:**
 - PTSD, anxiety, depression, anger management
 - Offender rehabilitation: reduce maladaptive behaviors
 - Physiotherapy: improve compliance, coping with pain or injury

2. Eye Movement Desensitization & Reprocessing (EMDR)

- **Definition:** Therapy designed to **process traumatic memories** and reduce distress
- **How it Works:**
 - Patient recalls trauma while performing **bilateral stimulation** (e.g., eye movements)
 - Helps **reprocess traumatic memories** and reduce emotional intensity
- **Applications:**
 - PTSD in victims and offenders
 - Trauma-informed rehabilitation alongside physiotherapy
 - Supports functional and emotional recovery

Key Takeaways

- **CBT:** Focuses on **thoughts, behaviors, coping**
- **EMDR:** Focuses on **processing traumatic memories**
- Both interventions can be integrated with **physiotherapy and criminology rehabilitation programs** for holistic care